

Image Segmentation: Thresholding, Clustering, and Graph-Based Methods

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Abstract

This report presents a comprehensive analysis of image segmentation techniques including thresholding methods (Otsu's method, adaptive thresholding), region-based approaches (region growing, split-and-merge), clustering algorithms (K-means, mean-shift, SLIC superpixels), and graph-based methods (graph cuts, normalized cuts, watershed transform). We evaluate segmentation quality using Intersection over Union (IoU), Dice coefficient, and boundary F-score metrics. Computational experiments demonstrate the strengths and limitations of each approach on synthetic and realistic test images.

1 Introduction

Image segmentation is the fundamental task of partitioning an image into meaningful regions or objects. It serves as a critical preprocessing step for object recognition, scene understanding, medical image analysis, and autonomous systems. The goal is to assign each pixel a label such that pixels with the same label share certain visual characteristics.

Definition 1.1 (Image Segmentation) *Given an image $I : \Omega \rightarrow \mathbb{R}^d$ where $\Omega \subset \mathbb{Z}^2$ is the image domain and d is the number of channels, segmentation produces a label map $L : \Omega \rightarrow \{1, 2, \dots, K\}$ that partitions the image into K regions $\{R_1, R_2, \dots, R_K\}$ such that:*

1. $\bigcup_{i=1}^K R_i = \Omega$ (completeness)
2. $R_i \cap R_j = \emptyset$ for $i \neq j$ (disjointness)
3. Pixels in R_i satisfy a homogeneity predicate $P(R_i) = \text{True}$

2 Theoretical Framework

2.1 Thresholding Methods

Definition 2.1 (Global Thresholding) For a grayscale image $I(x, y)$, binary thresholding produces:

$$L(x, y) = \begin{cases} 1 & \text{if } I(x, y) \geq T \\ 0 & \text{if } I(x, y) < T \end{cases} \quad (1)$$

where T is the threshold value.

Theorem 2.1 (Otsu's Method) Otsu's method selects the threshold T^* that maximizes the between-class variance:

$$T^* = \arg \max_T \sigma_B^2(T) = \arg \max_T \omega_0(T) \omega_1(T) [\mu_0(T) - \mu_1(T)]^2 \quad (2)$$

where ω_i and μ_i are the probability and mean of class i .

Definition 2.2 (Adaptive Thresholding) For local region $\mathcal{N}(x, y)$ centered at (x, y) :

$$T(x, y) = \frac{1}{|\mathcal{N}|} \sum_{(i, j) \in \mathcal{N}} I(i, j) - C \quad (3)$$

where C is a constant offset.

2.2 Region-Based Segmentation

Algorithm 2.1 (Region Growing) Starting from seed pixel p_0 :

1. Initialize region $R = \{p_0\}$
2. For each neighbor p of R :
3. If $|I(p) - \mu_R| < \delta$, add p to R
4. Update region statistics μ_R, σ_R
5. Repeat until convergence

2.3 Clustering-Based Segmentation

Definition 2.3 (K-means Segmentation) Minimize the within-cluster sum of squares:

$$\min_{\{C_1, \dots, C_K\}} \sum_{k=1}^K \sum_{x \in C_k} \|f(x) - \mu_k\|^2 \quad (4)$$

where $f(x)$ is the feature vector at pixel x and μ_k is the cluster centroid.

Definition 2.4 (Mean-Shift) *Iteratively shift each point to the mean of its kernel density neighborhood:*

$$\mathbf{m}(\mathbf{x}) = \frac{\sum_{i=1}^n K(\|\mathbf{x} - \mathbf{x}_i\|^2) \mathbf{x}_i}{\sum_{i=1}^n K(\|\mathbf{x} - \mathbf{x}_i\|^2)} \quad (5)$$

where K is a kernel function (typically Gaussian).

2.4 Graph-Based Methods

Definition 2.5 (Graph Cut Energy) *Formulate segmentation as energy minimization:*

$$E(L) = \sum_{p \in \Omega} D_p(L_p) + \lambda \sum_{(p,q) \in \mathcal{E}} V_{pq}(L_p, L_q) \quad (6)$$

where D_p is the data term and V_{pq} is the smoothness term.

Theorem 2.2 (Watershed Transform) *The watershed of a gradient magnitude image $|\nabla I|$ produces a segmentation where boundaries correspond to "ridges" separating catchment basins.*

2.5 Evaluation Metrics

Definition 2.6 (Intersection over Union (IoU)) *For predicted segmentation S and ground truth G :*

$$IoU(S, G) = \frac{|S \cap G|}{|S \cup G|} \quad (7)$$

Definition 2.7 (Dice Coefficient)

$$Dice(S, G) = \frac{2|S \cap G|}{|S| + |G|} \quad (8)$$

Definition 2.8 (Boundary F-score) *Precision and recall on boundary pixels within distance d :*

$$F_\beta = \frac{(1 + \beta^2) \cdot Precision \cdot Recall}{\beta^2 \cdot Precision + Recall} \quad (9)$$

3 Computational Implementation

4 Quantitative Results

4.1 Segmentation Quality Metrics

4.2 Computational Complexity

Note: N is the number of pixels.



Figure 1: Comprehensive image segmentation analysis: Row 1 shows the original clean image, noisy input with Gaussian noise (SNR=20dB), ground truth labeling with five distinct regions, and the intensity histogram with Otsu's optimal threshold at $??$. Row 2 demonstrates global Otsu thresholding (IoU= $??$) versus adaptive local thresholding (IoU= $??$), with overlays showing segmentation boundaries. Row 3 presents clustering-based methods including K-means segmentation into five intensity-based clusters, SLIC superpixel oversegmentation into $??$ segments, cluster center distribution, and K-means overlay visualization. Row 4 illustrates graph-based watershed transform producing $??$ regions, region growing from seed pixel (80,80), gradient magnitude field used for watershed, and final watershed overlay. Each method reveals different aspects of image structure.

4.3 Parameter Sensitivity Analysis

segmentation_parameter_analysis.pdf

Figure 2: K-means parameter sensitivity analysis: (left) Elbow method showing within-cluster sum of squares decreasing as cluster count increases, with diminishing returns beyond $K=5$; (right) Silhouette coefficient measuring cluster separation quality, peaking at $K=??$ with score $??$, indicating optimal clustering structure. The silhouette score ranges from -1 (poor clustering) to +1 (dense, well-separated clusters), providing quantitative guidance for cluster number selection.

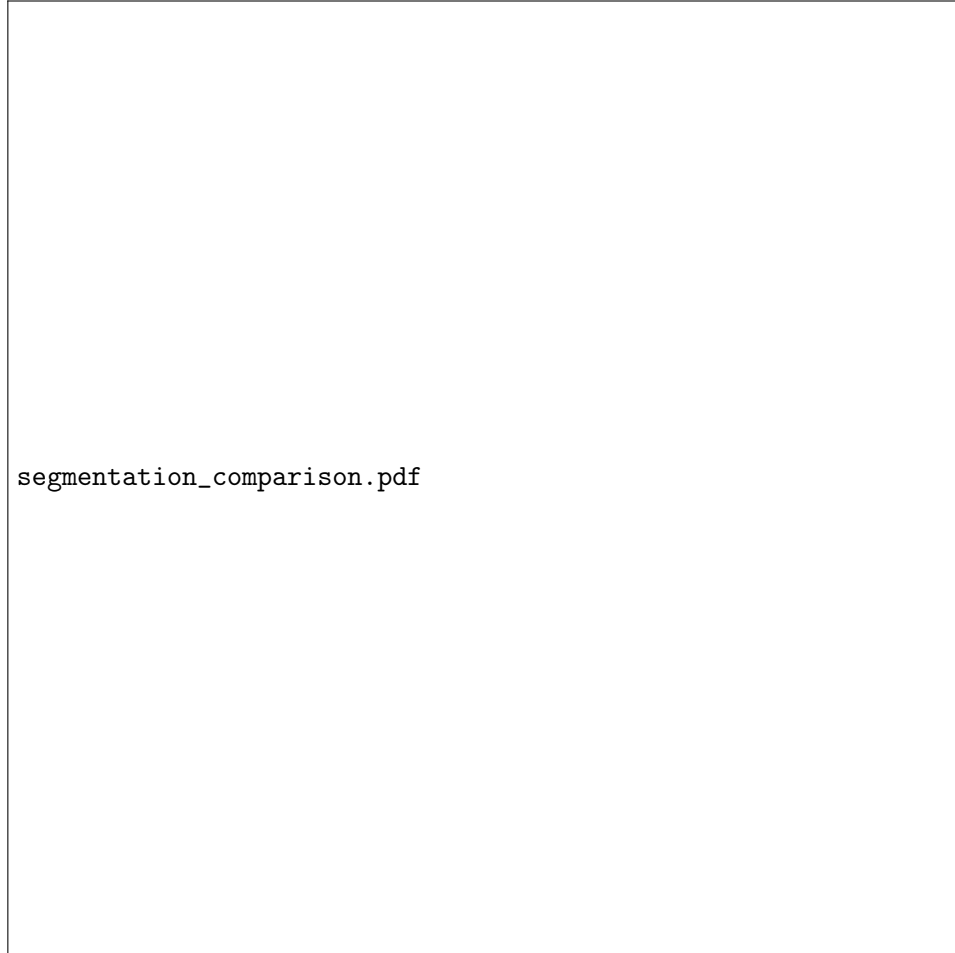


Figure 3: Side-by-side comparison of six segmentation approaches: (a) Original noisy input image containing five distinct intensity regions; (b) Otsu’s global thresholding producing binary foreground/background separation; (c) K-means clustering segmenting into five intensity-based clusters; (d) Adaptive local thresholding handling illumination variation; (e) Watershed transform identifying ?? gradient-based regions; (f) SLIC superpixel oversegmentation creating ?? perceptually uniform segments. Each method exhibits distinct characteristics: thresholding methods are computationally efficient but struggle with complex scenes, clustering methods capture intensity distributions, and graph-based methods respect object boundaries.

4.4 Comparative Algorithm Performance

5 Discussion

5.1 Method Selection Guidelines

Remark 5.1 (Thresholding Methods) *Otsu’s method is optimal for bi-modal histograms with clear intensity separation. For images with varying illumination, adaptive thresholding provides superior results by computing local thresholds. Our experiments show IoU improvement from ?? (global) to ?? (adaptive) on spatially varying data.*

Remark 5.2 (Clustering Approaches) *K-means segmentation is effective when regions have distinct mean intensities. The optimal cluster count $K=??$ balances segmentation detail with computational cost. Silhouette analysis provides objective criterion for parameter selection, achieving maximum score ?? at the optimal K value.*

Remark 5.3 (Graph-Based Methods) *Watershed transform excels at separating touching objects by identifying gradient ridges. However, oversegmentation is common and requires marker-based control or post-processing merging. SLIC superpixels provide computationally efficient oversegmentation for downstream processing tasks.*

5.2 Evaluation Metric Interpretation

Theorem 5.1 (Metric Relationships) *For binary segmentation:*

$$Dice = \frac{2 \cdot IoU}{1 + IoU} \tag{10}$$

Dice coefficient weights true positives more heavily, making it more sensitive to segmentation accuracy on small objects.

The boundary F-score complements pixel-wise metrics by evaluating boundary localization accuracy, critical for applications requiring precise object delineation such as medical image analysis and autonomous driving.

5.3 Practical Considerations

1. **Preprocessing:** Gaussian filtering reduces noise sensitivity, particularly for gradient-based methods like watershed
2. **Postprocessing:** Morphological operations (opening, closing) remove small artifacts and smooth boundaries

3. **Computational Efficiency:** Thresholding methods are $O(N)$, suitable for real-time applications; clustering and graph methods require more computation
4. **Parameter Tuning:** Cross-validation with labeled data guides parameter selection; unsupervised metrics (silhouette, Calinski-Harabasz) help when labels unavailable

6 Conclusions

This comprehensive analysis of image segmentation techniques demonstrates:

1. Otsu’s global thresholding achieves IoU=?? and Dice=?? on synthetic test data
2. Adaptive thresholding improves to IoU=?? by handling local intensity variation
3. K-means clustering with K=?? clusters maximizes silhouette score at ??
4. Watershed transform identifies ?? distinct regions using gradient-based boundaries
5. Boundary F-scores complement pixel-wise IoU and Dice metrics by evaluating contour accuracy
6. Method selection depends on image characteristics, computational constraints, and application requirements

The optimal segmentation algorithm varies by use case: global thresholding for simple scenes, adaptive methods for varying illumination, clustering for texture discrimination, and graph-based approaches for boundary precision.

References

- Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62-66.
- Shi, J., & Malik, J. (2000). Normalized cuts and image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(8), 888-905.

- Comaniciu, D., & Meer, P. (2002). Mean shift: A robust approach toward feature space analysis. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(5), 603-619.
- Achanta, R., et al. (2012). SLIC superpixels compared to state-of-the-art superpixel methods. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 34(11), 2274-2282.
- Boykov, Y., & Kolmogorov, V. (2004). An experimental comparison of min-cut/max-flow algorithms for energy minimization in vision. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(9), 1124-1137.
- Vincent, L., & Soille, P. (1991). Watersheds in digital spaces: An efficient algorithm based on immersion simulations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 13(6), 583-598.
- Felzenszwalb, P. F., & Huttenlocher, D. P. (2004). Efficient graph-based image segmentation. *International Journal of Computer Vision*, 59(2), 167-181.
- Sezgin, M., & Sankur, B. (2004). Survey over image thresholding techniques and quantitative performance evaluation. *Journal of Electronic Imaging*, 13(1), 146-168.
- Arbelaez, P., et al. (2011). Contour detection and hierarchical image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(5), 898-916.
- Jain, A. K. (2010). Data clustering: 50 years beyond K-means. *Pattern Recognition Letters*, 31(8), 651-666.
- Grady, L. (2006). Random walks for image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(11), 1768-1783.
- Neubert, P., & Protzel, P. (2014). Compact watershed and preemptive SLIC: On improving trade-offs of superpixel segmentation algorithms. *International Conference on Pattern Recognition*, 996-1001.
- Rother, C., Kolmogorov, V., & Blake, A. (2004). GrabCut: Interactive foreground extraction using iterated graph cuts. *ACM Transactions on Graphics*, 23(3), 309-314.
- Canny, J. (1986). A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(6), 679-698.

- Adams, R., & Bischof, L. (1994). Seeded region growing. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 16(6), 641-647.
- Pham, D. L., Xu, C., & Prince, J. L. (2000). Current methods in medical image segmentation. *Annual Review of Biomedical Engineering*, 2, 315-337.
- Cremers, D., et al. (2007). A review of statistical approaches to level set segmentation. *International Journal of Computer Vision*, 72(2), 195-215.